Topic 5 — Agriculture and Indian Agricultural Economy

> **Sources baked in:** Ramesh Singh Ch.8, Sanjiv Verma, Agricultural Census 2015-16, Economic Survey, ICAR handbooks, UPPCS Prelims PYQs; Extra Drill rebuilt from mixed RO/ARO-coaching stems (paste full Ghatnachakra Purvalokan later to expand) **Weight:** ★★★ — holdings, seasons, irrigation, fertilizer subsidy, NABARD/KCC, PMFBY, WTO boxes **Last verified:** September 2026 — Lucent remodel (History/Polity section order)

MSP / revolutions / NFSA scheme depth → Topic 6. This chapter owns agrarian structure, crops, inputs, credit and WTO AoA desk.

Current Affairs

Year	Fact	Why it matters
IYM / Shree Anna	UN International Year of Millets **2023**; India brands millets as Shree Anna.	Millet producer / export stems.
Nano urea / DAP	IFFCO Kalol commercial nano liquid urea / nano DAP plants.	Place–product identity.
PM-PRANAM	Union Budget 2023–24 incentive for states to cut chemical fertilizer use.	Scheme name stem.
Agri-Stack / DPI	Digital Public Infrastructure for Agriculture narrative.	Rolling CA — refresh figures from Survey.

5.1 Agrarian Structure and Operational Holdings

- **Primary Sector Contribution:** Agriculture, forestry, and fishing contribute approximately **17%–18% of India's Gross Value Added (GVA)** at current basic prices. Despite its declining economic share in GDP (from ~54% in 1950–51), it remains the backbone of the rural economy, directly employing **~45%–47% of the Indian workforce** according to the Periodic Labour Force Survey (PLFS).
- **The Land Holding Paradigm:** The structure of land ownership in India is characterised by acute **fragmentation** and dominance of small, economically unviable plots.
- **Agricultural Census Findings (2015–16):**
- **Marginal Farmers (< 1 ha):** Constitute **68.5%** of all operational holdings (operating ~24.0% of area).
- **Small Farmers (1–2 ha):** Constitute **17.7%** of holdings (operating ~23.0% of area).
- **Combined Small & Marginal Farmers (SMFs, < 2 ha):** Constitute **86.2%** of all holdings in India, controlling **47.3%** of total operated area.
- **Medium & Large Farmers (> 4 ha):** Make up less than **4.4%** of holdings, but control over **30.9%** of operated land.

- **Average Operational Size:** Declined from **2.28 ha in 1970–71** to **1.15 ha in 2010–11**, and further to **1.08 ha in 2015–16**.
- **Economic Implications of Fragmentation:** 1. Prevents economies of scale in farm operations. 2. Severely limits mechanisation (e.g. combine harvesters cannot navigate fragmented strips). 3. High transaction costs for institutional credit delivery. 4. Farmers remain vulnerable to climate volatility, price shocks, and local moneylenders.

> **Logic:** Remember the cut-off thresholds for UPPSC: Marginal is < 1 hectare; Small is 1–2 hectares; Semi-Medium is 2–4 hectares; Medium is 4–10 hectares; Large is ≥ 10 hectares. Over 86% of Indian farmers are Small or Marginal!

Inline PYQ

In the Agricultural Census of India, a farmer operating an agricultural holding of size between 1.00 hectare and 2.00 hectares is designated as:

A. Marginal Farmer
B. Small Farmer
C. Semi-Medium Farmer
D. Medium Farmer

Show answer

Logic: Marginal is below 1 hectare (< 1.0 ha), Small is 1 to 2 hectares (1.0–2.0 ha), Semi-Medium is 2 to 4 hectares, Medium is 4 to 10 hectares, and Large is 10 hectares and above.

Ans: B. Option B** is correct: — Small Farmer..

5.2 Cropping Patterns & Major Crops in India

- **Cropping Pattern (शस्य प्रतिरूप):** Refers to the proportion of area under different crops at a given point in time in a region.
- In India, **food grains (खाद्यान्न)** occupy approximately **65%–68% of the gross cropped area**, reflecting the subsistence orientation of the agrarian structure, while **commercial/cash crops (नकदी फसलें)** occupy ~32%–35%.
- **Gross Cropped Area (GCA) vs Net Sown Area (NSA):**
- **Net Sown Area (NSA):** The total physical land area sown with crops at least once in an agricultural year (~139–140 million hectares in India).
- **Gross Cropped Area (GCA):** Total area sown once as well as more than once in the year (~198–200 million hectares).
- **Cropping Intensity (शस्य सघनता):** Calculated as: > **Formula:** `Cropping Intensity = (Gross Cropped Area / Net Sown Area) × 100` India's average cropping intensity is approximately **~142%–143%**. **Punjab** records the highest cropping intensity (>190%), while rainfed dryland states record lower ratios (~115%–125%).

Detailed Crop Profiles & Leading State Producers

1. Rice (चावल / Paddy - *Oryza sativa*)

- **Status:** Primary staple food crop of India; occupies the largest gross cropped area (~47 million hectares).
- **Agro-climatic Needs:** Warm and humid tropical climate; mean temperature 22°C – 32°C; annual rainfall 100 cm – 200 cm; deep clayey or loamy alluvial soils with water-retaining capacity.
- **Leading Producers: West Bengal (1st)**, **Uttar Pradesh (2nd)**, **Punjab (3rd)**.
- **Productivity Champion: Punjab** records the highest yield per hectare (>4,300 kg/ha) due to 100% canal and tube-well irrigation and high fertilizer consumption.
- **Aus, Aman, Boro:** In West Bengal, Assam, and Odisha, three seasonal rice crops are harvested in a single year: **Aus** (autumn rice, sown May/June), **Aman** (winter rice, sown June/July), and **Boro** (summer rice, sown Nov/Dec).

2. Wheat (गेहूं - *Triticum aestivum*)

- **Status:** Second most crucial cereal crop; backbone of northern India's food security and public distribution system.
- **Agro-climatic Needs:** Temperate rabi crop; cool growing period (10°C – 15°C) and warm sunny weather during ripening (20°C – 25°C); rainfall 50 cm – 75 cm; well-drained fertile loamy soils.
- **Western Disturbances (पश्चिमी विक्षोभ):** Light winter cyclonic rainfall in December–January across Punjab, Haryana, and Western UP is extremely beneficial for wheat tillering and grain filling.
- **Leading Producers: Uttar Pradesh (1st in absolute output ~32%–34% of national total)**, **Madhya Pradesh (2nd)**, **Punjab (3rd)**.
- **Productivity Champion: Punjab and Haryana** lead in yield (>4,800 kg/ha).

3. Shree Anna / Millets (मोटे अनाज)

- **Crops: Bajra (Pearl Millet)**, **Jowar (Sorghum)**, **Ragi (Finger Millet)**, Small millets (Kodo, Kutki, Sanwa, Cheena).
- **Nutritional & Climate Significance:** Highly drought-resistant, climate-resilient C4 plants requiring minimal irrigation and chemical fertilizers; rich in dietary fiber, iron, calcium, and micronutrients.
- **Policy Push:** Championed by India, the UN General Assembly declared **2023 as the International Year of Millets (IYM)**. The Government rebranded millets as **"Shree Anna"** and designated ICAR-Indian Institute of Millets Research (IIMR), Hyderabad as a Global Centre of Excellence.
- **Leading Producers:**
- **Bajra: Rajasthan (1st)**, Uttar Pradesh, Gujarat.
- **Jowar: Maharashtra (1st)**, Karnataka.
- **Ragi: Karnataka (1st)**.

4. Pulses (दलहन)

- **Status:** Vital source of plant protein in vegetarian diets; leguminous crops that fix atmospheric nitrogen through *Rhizobium* root nodules, naturally restoring soil fertility.
- **Major Crops: Gram (चना - ~45% of total pulses)**, **Tur/Arhar (अरहर/तूर)**, **Urad (उड़द)**, **Moong (मूंग)**, **Masoor (मसूर)**.
- **Leading Producers: Madhya Pradesh (1st)**, **Rajasthan (2nd)**, **Maharashtra (3rd)**.

- **Economic Paradox:** India is the **world's largest producer (25%)**, **largest consumer (27%)**, and **largest importer (14%)** of pulses.

5. Oilseeds & Cash Crops

- **Oilseeds (तिलहन):** India meets ~55%–60% of its edible oil consumption through imports (predominantly Palm oil from Indonesia/Malaysia and Soybean/Sunflower oil from South America/Ukraine).
- **Soybean: Madhya Pradesh & Maharashtra** (Madhya Pradesh is called *Soya State*).
- **Groundnut: Gujarat (1st)**, Rajasthan.
- **Rapeseed & Mustard: Rajasthan (1st)**, Madhya Pradesh, Haryana.
- **Sugarcane (गन्ना): Uttar Pradesh (1st in area & production)**, Maharashtra, Karnataka.
- **Cotton (कपास): Gujarat (1st)**, Maharashtra, Telangana.
- **Jute (जूट - Golden Fibre): West Bengal (1st ~75%–80% of national total)**, Bihar, Assam.

Inline PYQ

Which one of the following states is the largest producer of Pulses in India?

A. Uttar Pradesh
B. Madhya Pradesh
C. Rajasthan
D. Maharashtra

Show answer

Logic: Madhya Pradesh consistently ranks first in total pulse production in India (accounting for over 25% of national output), followed by Rajasthan and Maharashtra.

Ans: B. Option B** is correct: — Madhya Pradesh..

5.3 Irrigation Architecture and Water Management

- **The Irrigation Imperative:** More than **50% of India's Net Sown Area is rainfed (बारानी कृषि)**, making agricultural output heavily sensitive to the spatial and temporal vagaries of the Indian Summer Monsoon.
- **Irrigation Status:** Gross Irrigated Area is approximately **102–103 million hectares**, while Net Irrigated Area is **~70–73 million hectares**.

Source-wise Share in Net Irrigated Area of India

Irrigation Source	National Share (%)	Geographical Distribution & Leading States	Economic Advantages & Environmental Hazards
Wells & Tube-wells (नलकूप एवं कुएं)	**~62% – 64%** (Dominant Source)	Heavily concentrated in the Indo-Gangetic Plains: **Uttar Pradesh, Punjab, Haryana, Rajasthan, Gujarat, Bihar**.	**Advantages:** Independent farmer control, on-demand irrigation, high application efficiency. **Hazards:** Over-exploitation of aquifers, creating "dark zones" and ground subsidence; power subsidy distortion.
Canals (नहरें)	**~23% – 25%** (Second Largest)	Extensive in alluvial river plains: **Uttar Pradesh, Punjab, Haryana, Andhra Pradesh, Rajasthan (Indira Gandhi Canal)**.	**Advantages:** Harnesses perennial river runoff; low recurring operational cost for farmers. **Hazards:** Waterlogging (*Seep*), soil salinization and alkalinization (*Reh/Kallar*), siltation of canals.
Tanks (तालाब / जलाशय)	**~3% – 4%**	Ancient indigenous system prevalent in the hard-rock Deccan plateau: **Tamil Nadu, Andhra Pradesh, Telangana, Karnataka**.	**Advantages:** Eco-friendly, rainwater harvesting, recharges groundwater. **Hazards:** Silt accumulation, highly dependent on local seasonal rainfall, seasonal drying.
Other Sources (Lifts, Springs, Streams)	**~7% – 9%**	Mountainous valleys, coastal deltas, wetlands: **Uttarakhand, Himachal, North-Eastern states**.	Localized small-scale mountain systems (*Kuhl* in Himachal).

Flagship Irrigation Schemes

1. Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)

- **Launched: July 2015** with an overarching objective to expand cultivable area under assured irrigation and improve on-farm water use efficiency.
- **Motto:** *"Har Khet Ko Pani"* (water to every field) and *"More Crop Per Drop"* (enhancing water productivity).
- **Core Components:** 1. **Accelerated Irrigation Benefit Programme (AIBP):** Focuses on faster completion of ongoing major and medium irrigation projects. 2. **PMKSY - Har Khet Ko Pani:** Creating new water sources through minor irrigation (both surface and groundwater), repair, restoration and renovation (RRR) of traditional water bodies. 3. **PMKSY - Per Drop More Crop (PDMC):** Promotes micro-irrigation technologies (**Drip Irrigation** and **Sprinkler Irrigation**) to save water (up to 40%–50%) and reduce fertilizer wastage through fertigation. 4. **PMKSY - Watershed Development:** Ridge-to-valley soil and moisture conservation, check dams, and rainwater harvesting.

Inline PYQ

In India, which one among the following is the largest source of irrigation in terms of net irrigated area?

A. Canals
B. Tanks
C. Tube-wells and Wells
D. Others

Show answer

Logic: Tube-wells and open wells together account for ~62%–64% of the total net irrigated area of India, followed by canals (~23%–25%) and tanks (~3%–4%).

Ans: C. Option C** is correct: — Tube-wells and Wells..

5.4 Agricultural Inputs: Seeds, Fertilizers, and Mechanisation

A. Seeds (बीज) — The Fundamental Biological Catalyst

- Seed is the most vital input determining genetic crop productivity; **Seed Replacement Rate (SRR)** is the percentage of total cropped area sown with certified/quality seeds relative to farm-saved seeds.
- **Classes of Seeds in India:** 1. **Breeder Seed (प्रजनक बीज):** Produced under direct supervision of the originating plant breeder at ICAR institutes/State Agricultural Universities; 100% genetic purity; tagged with **Golden Yellow label**. 2. **Foundation Seed (आधार बीज):** Progeny of breeder seed produced by National Seeds Corporation (NSC) or State Seed Corporations; tagged with **White label**. 3. **Certified Seed (प्रमाणित बीज):** Progeny of foundation seed certified by State Seed Certification Agencies and sold directly to commercial farmers; tagged with **Azure Blue label**.

B. Chemical Fertilizers & Subsidy Mechanisms

- **Recommended NPK Ratio: 4 : 2 : 1** (N:P:K). Actual consumption in several high-yield states has skewed past **8:3:1** due to disproportionate urea subsidization.
- **1. Urea Subsidy Regime:**
- Urea is the **only fertilizer remaining under statutory price control (fixed statutory MRP)**.
- The difference between the delivered cost of production of urea and its government-notified retail price is disbursed directly to urea manufacturers as a subsidy under the **New Urea Policy (NUP)**.
- **100% Mandatory Neem Coated Urea (NCU, 2015):** Mandatory coating with neem oil slows nitrification in soil, preventing groundwater leaching and eliminating illegal diversion of subsidized urea to chemical, paint, and plywood industries.
- **2. Nutrient Based Subsidy (NBS) Scheme (2010):**
- Introduced in **April 2010** under the Department of Fertilizers for non-urea P&K fertilizers (**DAP, MOP, NPK complexes, Single Super Phosphate - SSP**).
- Central Government announces a **fixed annual per-kilogram subsidy** for each nutrient (N, P, K, and Sulphur S).
- **MRPs of P&K fertilizers are decontrolled** and determined by market dynamics and global raw material costs.

- **Potash Import Dependence:** India possesses **zero commercially extractable reserves of Potash** and is **100% dependent on imports** for Muriate of Potash (MOP).
- **3. Nano Fertilizers:**
- Developed and patented by IFFCO; **Nano Urea (Liquid)** and **Nano DAP**.
- A 500 ml bottle replaces an entire 45 kg bag of conventional granular urea, cutting logistics costs, reducing soil acidification, and raising nutrient use efficiency from ~30% to over ~80%.

> **Traps:** Urea = statutorily controlled MRP; P&K (DAP, MOP) = decontrolled MRP under NBS. India has 100% import dependence for Potash (MOP)!

Inline PYQ

Regarding the Nutrient Based Subsidy (NBS) scheme for fertilizers in India, consider the following:

1. It applies to all chemical fertilizers including Urea.
2. Under NBS, the MRP of non-urea P&K fertilizers is determined by fertilizer companies.

Which of the statements given above is/are correct?

A. 1 only
B. 2 only
C. Both 1 and 2
D. Neither 1 nor 2

Show answer

Logic: Statement 1 is incorrect because Urea is strictly excluded from NBS and remains under direct statutory price control.

Ans: B. Option B** is correct: — 2 only..

5.5 Sustainable, Organic and Natural Farming

1. Organic Farming (जैविक कृषि)

- Agricultural practice that avoids synthetic chemical fertilizers, chemical pesticides, growth regulators, and genetically modified organisms (GMOs).
- Relies on crop rotation, green manures, animal manures, bio-fertilizers (*Azotobacter, Rhizobium, PSB*), and biological pest control.
- **Sikkim Landmark:** Formally conferred status as the **World's First 100% Organic State** by the UN FAO in **2016** (Future Policy Gold Award).
- **Paramparagat Krishi Vikas Yojana (PKVY):** Launched in **2015** to promote cluster-based organic farming through **Participatory Guarantee System (PGS-India)** certification without incurring heavy third-party audit fees.

2. Natural Farming vs Organic Farming

- **Organic farming** often permits off-farm organic inputs (e.g. commercially purchased vermicompost, bio-fertilizers) and requires rigorous certification.
- **Natural farming** strictly prohibits all external purchased inputs; all soil nutrients and pest repellents must be prepared on-farm using indigenous cow dung, cow urine, and local biomass.
- **Zero Budget Natural Farming (ZBNF):**
- Pioneered by Padma Shri **Subhash Palekar** in Maharashtra and Karnataka.
- **Four Pillars of ZBNF:**
  1. **Jeevamrutha (जीवामृत):** Fermented liquid microbial inoculant made of fresh indigenous cow dung, cow urine, jaggery, pulse flour, virgin soil, and water; stimulates soil earthworms and microbial activity.
  2. **Bijamrutha (बीजामृत):** Seed treatment concoction using cow dung, urine, and lime to protect young seedlings from seed-borne and soil-borne fungal pathogens.
  3. **Acchadana (आच्छादन - Mulching):** Soil, straw, and live mulching to preserve topsoil moisture and suppress weeds.
  4. **Whapasa (वापसा):** Micro-climatic soil moisture aeration condition where both air and water molecules coexist in equal balance in root zones, slashing water needs by over 80%.

5.6 Agricultural Marketing, Mandis and Supply Chains

- **The Traditional APMC Structure:**
- Agricultural marketing is a **State subject** under Entry 28 of List II (State List) of the Seventh Schedule.
- Historically, state governments established **Agricultural Produce Market Committees (APMCs)** to protect farmers from exploitation by middlemen and moneylenders.
- **APMC Market Failures:** Over time, APMC mandis degenerated into cartels, charging steep mandi fees, multiple levies, licensing barriers for new traders, lack of transparent electronic bidding, and high post-harvest losses.
- **Model APMC Act (2003) & APLM Act (2017):**
- Drafted by the Central Government to nudge states to allow private wholesale markets, direct purchase from farmers, single-point levy of market fee, and promotion of public-private partnerships.
- **Electronic National Agriculture Market (e-NAM):**
- **Launched: 14 April 2016** by Prime Minister Narendra Modi.
- **Core Architecture:** A virtual electronic trading platform that digitally integrates existing physical APMC mandis across India to create a unified national agricultural market.
- **Nodal Implementing Agency: Small Farmers' Agri-Business Consortium (SFAC)** under the Ministry of Agriculture.
- **Benefits:** Online transparent competitive bidding, assaying and quality certification facilities at mandi gate, electronic online payments directly into farmers' bank accounts, eliminating local commission agent cartels.

- **Farmer Producer Organisations (FPOs):**
- Institutional aggregation of small and marginal farmers into producer companies under the Companies Act, 2013.
- Enhances smallholder bargaining power in purchasing farm inputs at wholesale rates and selling produce directly to institutional buyers and exporters.
- Central Sector Scheme: **Formation and Promotion of 10,000 FPOs** launched in **2020** with a dedicated credit guarantee fund in NABARD.

Inline PYQ

The electronic trading portal 'e-NAM' (National Agriculture Market) is implemented by which of the following agencies?

A. National Agricultural Cooperative Marketing Federation of India (NAFED)
B. Small Farmers' Agri-Business Consortium (SFAC)
C. Commission for Agricultural Costs and Prices (CACP)
D. Food Corporation of India (FCI)

Show answer

Logic: e-NAM was launched on 14 April 2016 as a pan-India electronic trading portal networking APMC mandis.

Ans: B. Option B** is correct: — Small Farmers' Agri-Business Consortium (SFAC)..

5.7 Institutional Agricultural Credit and Rural Finance

- **Historical Perspective:** At Independence, non-institutional informal sources (village moneylenders, traders, zamindars) supplied over **93% of all rural credit** at usurious interest rates (All-India Rural Credit Survey Committee, 1954).
- Today, **formal institutional sources supply over 75%** of agricultural credit, while informal sources account for under 25%.

Institutional Lenders Profile

1. **Commercial Banks (Public and Private Banks):** - Provide the lion's share (**~75%–78%**) of total institutional farm credit disbursed annually. - Excel in disbursing agricultural term loans (for tractors, solar pumps, cold storage).
2. **Regional Rural Banks (RRBs - क्षेत्रीय ग्रामीण बैंक):** - Established under the **RRB Act, 1976** based on the recommendations of the **M. Narasimham Working Group (1975)**. - First RRB: **Prathama Bank** (Moradabad, UP, sponsored by Syndicate Bank on 2 October 1975). - Equity ownership: **50% Central Govt : 15% State Govt : 35% Sponsor Bank**. - Account for **~12%–14%** of agricultural credit.
3. **Cooperative Credit Institutions (सहकारी ऋण संस्थाएं):** - Short-term rural cooperative credit structure is three-tiered:
    1. **PACS (Primary Agricultural Credit Societies)** at the village level.
    2. **DCCB (District Central Cooperative Banks)** at the district level.

3. **StCB (State Cooperative Banks)** at the apex state level. - Account for **~10%–12%** of agricultural credit, but hold the largest rural customer footprint among marginal farmers.

Apex Institutions and Priority Lending Mandates

1. NABARD (राष्ट्रीय कृषि और ग्रामीण विकास बैंक)

- **Established: 12 July 1982** under the NABARD Act, 1981, upon the recommendations of the **B. Sivaraman Committee (CRAFICARD)**.
- **Mandate:** Apex development and refinancing bank for agriculture, small-scale industries, cottage and village industries, handicrafts, and rural crafts.
- **Equity:** 100% owned by the Government of India (RBI transferred its share to the Govt).
- **Core Facilities:**
- Refinances Commercial Banks, RRBs, and State Cooperative Banks for short-term and long-term agricultural lending.
- Manages the **Rural Infrastructure Development Fund (RIDF)** set up in **1995–96**.
- Operates the **Long Term Irrigation Fund (LTIF)** and **Micro Irrigation Fund (MIF)**.

2. Kisan Credit Card (KCC) Scheme

- **Introduced: August 1998** by NABARD and RBI on the recommendations of the **R.V. Gupta Committee**.
- **Objective:** Provide adequate, timely, and cost-effective credit to farmers under a single window for: 1. Short-term crop cultivation expenses. 2. Post-harvest expenses and produce marketing loans. 3. Working capital for maintenance of farm assets and allied activities. 4. Consumption requirements of farmer households (up to 10%).
- **Interest Subvention Scheme (ISS) / Modified ISS:**
- Benchmark short-term crop loan up to **₹3 lakh** is provided at a subsidized base interest rate of **7%**.
- Farmers ensuring **prompt repayment** receive an additional **3% prompt repayment incentive (PRI)**, bringing the effective interest rate down to just **4% per annum**.
- **Expanded Scope (2018–19):** KCC facility extended to **Animal Husbandry, Dairy, and Fisheries** farmers for working capital needs up to ₹2 lakh at similar concessional interest rates.

Inline PYQ

The Kisan Credit Card (KCC) scheme was introduced in India in which year?

A. 1991
B. 1995
C. 1998
D. 2001

Show answer

Logic: The Kisan Credit Card (KCC) scheme was formulated by NABARD and introduced by the Government of India in August 1998 on the recommendations of the R.V.

Ans: C. Option C** is correct: — 1998..

5.8 Agricultural Risk, Insurance & Allied Economy

A. Pradhan Mantri Fasal Bima Yojana (PMFBY)

- **Launched: February 2016** (operational from Kharif 2016 season) under the Ministry of Agriculture.
- **Replaced:** National Agricultural Insurance Scheme (NAIS) and Modified NAIS (MNAIS).
- **Uniform Low Premium Paid by Farmers:**
- **2.0%** of sum insured for all **Kharif food and oilseed crops**.
- **1.5%** of sum insured for all **Rabi food and oilseed crops**.
- **5.0%** of sum insured for all **Annual Commercial and Horticultural crops**.
- **Subsidy Sharing:** The difference between actuarial premium rates and farmer's premium is shared **50:50 between the Central and State Governments** (90:10 for North-Eastern States).
- **Key Features:**
- Comprehensive risk coverage: prevented sowing/planting risk, localized calamities (landslide, hailstorm, inundation), mid-season adversity, and post-harvest losses (up to 14 days for crops kept in "cut-and-spread" condition in the field).
- Uses modern remote sensing, drones, and satellite imagery for rapid Crop Cutting Experiments (CCEs) and direct DBT claim settlements.
- Made **voluntary for all farmers** (including loanee farmers) since Kharif 2020.

B. Allied Agricultural Sectors

1. Animal Husbandry & Dairy (पशुपालन एवं डेयरी)

- India is the **world's largest milk producer**, contributing ~24% of global milk output (~230 million tonnes in 2022–23).
- **Operation Flood (ऑपरेशन फ्लड):** Launched in **1970** by the National Dairy Development Board (NDDB) headed by **Dr. Verghese Kurien**; established the three-tier cooperative milk federation pattern (Anand model / AMUL).
- **Rashtriya Gokul Mission (2014):** Focused on indigenous bovine breed conservation and productivity enhancement.

2. Fisheries Sector & Blue Revolution (नीली क्रांति)

- India is the **second-largest fish producing country** (after China), contributing ~8% to global fish production.
- **Pradhan Mantri Matsya Sampada Yojana (PMMSY):** Launched in **May 2020** with an outlay of ₹20,050 crore to modernize fish infrastructure, harbor landing centers, cage culture, and boost seafood exports.

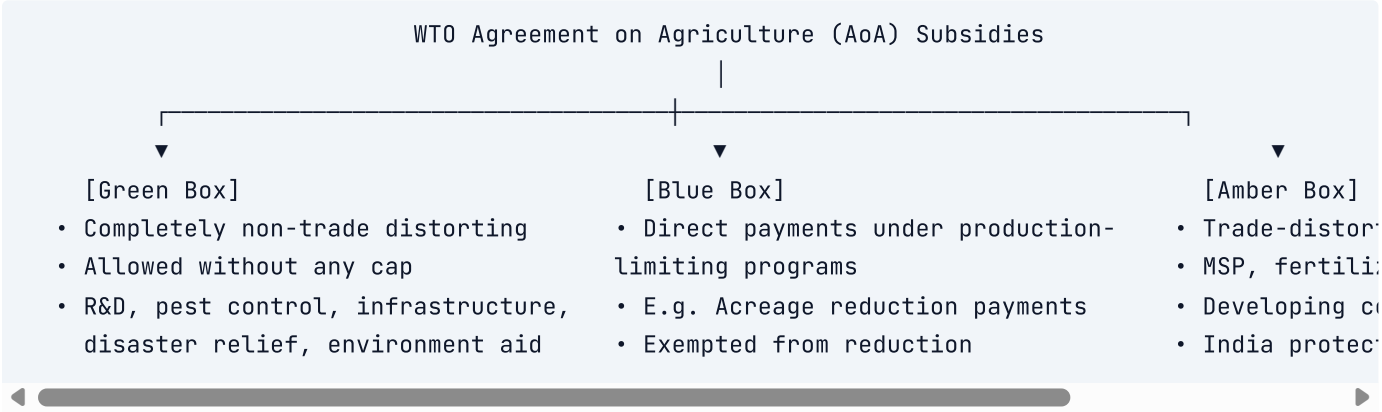
5.9 Structural Challenges & WTO Trade Regimes

A. Disguised Unemployment in Agriculture (प्रच्छन्न बेरोजगारी)

- A socio-economic condition where **more people are engaged in an economic activity than fundamentally necessary**.
- If surplus laborers are withdrawn from the agricultural plot, the **total physical agricultural output remains completely unchanged**.
- Therefore, the **marginal physical productivity of surplus farm labor is ZERO (or negative)**.
- Caused by lack of alternative non-farm rural industrial employment, high population growth, and joint-family tenure systems.

B. WTO Agreement on Agriculture (AoA) — The Three Subsidy Boxes

- Negotiated during the Uruguay Round (1986–1994) to reduce agricultural trade distortions and export subsidies.



1. **Green Box (हरा बॉक्स):** - Subsidies that cause **zero or minimal trade-distorting effects**. - Must be publicly funded and not involve price support. - Examples: Agricultural research, extension training, pest control, rural infrastructure, direct decoupled income transfers, disaster relief payments, and environmental conservation programs. - **Status: Allowed without any financial ceiling**.
2. **Blue Box (नीला बॉक्स):** - Direct government payments to farmers under **production-limiting programs** (based on fixed area and yields or a fixed number of livestock). - Designed to avoid market glut. - **Status: Exempted from reduction commitments**.
3. **Amber Box (अंबर बॉक्स):** - All domestic support measures that directly **distort international production and trade** by affecting market prices. - Examples: **Minimum Support Prices (MSP)**, input subsidies on fertilizers, canal water, electricity, and seeds. - **Ceilings (*De-minimis* Limits):**
   - Developed nations: Capped at **5%** of their total value of agricultural production.
   - Developing nations (India): Capped at **10%** of their total agricultural production value.
4. **The Peace Clause & Bali Ministerial Conference (2013):** - India successfully negotiated the **Peace Clause** at the 9th WTO Ministerial Conference in Bali (2013), and made it permanent in 2014. - **Mandate:** Protects developing countries from being challenged or penalized under WTO dispute settlement mechanisms if their domestic food security procurement programs (such as MSP-backed procurement for the National Food Security Act, 2013) breach the 10% Amber Box *de-minimis* threshold.

Inline PYQ

In the context of the World Trade Organization (WTO), 'Amber Box, Blue Box and Green Box' subsidies are related to:

A. Industrial Manufacturing
B. Agricultural Affairs
C. Banking and Financial Services
D. Information and Communication Technology

Show answer

Logic: Under the WTO Agreement on Agriculture (AoA), domestic agricultural support programs and subsidies are classified into three colored boxes: Green Box (non-distorting), Blue Box (production-limiting direct aids), and A…

Ans: B. Option B** is correct: — Agricultural Affairs..